



HOUSE PRICE PREDICTION USING MACHINE LEARNING

- **MID-PROJECT REVIEW**

Machine Learning & Data Analytics

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1. PROGRESS SUMMARY

1.1 Project Objective and Dataset

This project develops machine learning models to predict median housing prices at the county level across the United States. The primary objectives are to achieve a Root Mean Squared Error (RMSE) below \$25,000 and a coefficient of determination (R^2) exceeding 0.85. These targets will demonstrate the model's ability to accurately forecast housing prices based on geographic, temporal, and market-driven factors.

The research addresses several critical questions: Can machine learning models effectively predict housing prices using publicly available data? Which features most significantly influence price variations? How do geographic and temporal factors impact housing market dynamics? The answers to these questions have practical applications for real estate investment, policy planning, and economic forecasting.

The dataset utilized is the Redfin Housing Market Data sourced from Kaggle, encompassing monthly county-level housing statistics from January 2012 through December 2021. This comprehensive dataset originally contained 563,122 records across 1,860 counties in 47 US states with 58 initial features including price metrics, sales volumes, inventory levels, days on market, and various market condition indicators. The temporal span of nearly 10 years provides sufficient data to capture market cycles, seasonal patterns, and the impact of major economic events including the COVID-19 pandemic.

1.2 Work Completed to Date

Significant progress has been achieved across four major project phases, representing approximately 70% of the total project timeline. The approach followed industry-standard data blending methodology from initial exploration through preliminary model development.

Phase 1: Exploratory Data Analysis

Comprehensive exploration revealed critical insights about the US housing market. Geographic analysis demonstrated substantial price variation, with median prices ranging from approximately \$125,000 in West Virginia to over \$550,000 in markets such as Hawaii, California, and the District of Columbia. This four-fold difference highlighted the importance of location-based features for predictive modeling. For this period, I have excluded NY due to large amount of data. I have planned to add that part as well in the final report.

Temporal analysis identified distinct market phases: steady recovery from the 2008 financial crisis (2012-2015), moderate growth (2016-2019), and unprecedented price acceleration during COVID-19 (2020-2021), with some markets experiencing 15-20% annual increases. Seasonal

pattern analysis revealed consistent cyclical trends, with spring and summer months showing prices 5-10% higher than fall and winter periods.

A total of 11 professional visualizations were created, including missing value analyses, price distributions, state-level comparisons, temporal trends, seasonal patterns, and property type breakdowns. These visualizations provided critical insights that informed subsequent preprocessing and feature engineering decisions.

Phase 2: Data Preprocessing

Data preprocessing addressed missing values, outliers, and data type conversions. A multi-step approach was implemented: rows with missing target variables were removed (1,685 records), columns with over 50% missing values were dropped, forward-fill was applied for time-series continuity, and median imputation handled remaining gaps. This strategy preserved temporal patterns while maximizing usable data.

Outlier analysis using the Interquartile Range (IQR) method initially identified approximately 0.5% of records as statistical outliers. However, after careful examination, these outliers were retained as they represented genuine geographic variation rather than data errors. Manhattan and San Francisco counties legitimately command prices 10 times higher than rural counties in less expensive states.

Categorical variables were encoded appropriately: one-hot encoding for property types (low cardinality), label encoding for state codes (high cardinality). Initial feature engineering created temporal features (year, month, quarter, season indicators), lag features (1-month, 3-month, 12-month lags), and basic calculated features (price-to-list ratios, inventory-to-sales ratios).

Phase 3: Predictive Variable Creation

Extensive feature engineering created 61 new predictive variables across six categories. Geographic features captured price hierarchies at state and county levels, including state median prices, price indices, and county rankings within states. Temporal rolling window features computed 3-month, 6-month, and 12-month moving averages, volatility measures, and momentum indicators.

Year-over-year growth metrics quantified annual price changes, sales volume changes, and inventory fluctuations. Market condition indicators measured supply-demand ratios, competition levels through sale-to-list price ratios, and market heat through days-on-market statistics. Seasonal indicators flagged peak and off-peak buying periods. Interaction features captured combined effects such as state price levels during peak seasons and market conditions under low supply.

The final engineered dataset contained 562,437 records with 119 features, expanding from the original 58 columns. This feature-rich dataset provided the foundation for preliminary model training.

Phase 4: Preliminary Model Training

Two baseline models were trained: **Linear Regression** and **Random Forest**. The modeling process revealed critical methodological challenges that strengthened the final approach. Initial

results showed suspiciously high performance ($R^2 = 0.9992$), leading to identification of data leakage issues. Features derived directly from the target variable, such as median sale price month-over-month changes and price-per-square-foot metrics, were inadvertently included, essentially allowing models to use the answer to predict the answer.

After removing leaking features, models initially performed poorly ($R^2 < 0.01$), revealing additional data quality issues. Extreme placeholder values (\$999,999,999) in the dataset were identified as missing data indicators. Aggressive data cleaning removed these placeholders and applied IQR-based outlier filtering, reducing the dataset to 548,629 clean records.

With cleaned data and legitimate features, preliminary models achieved realistic baseline performance. Linear Regression attained RMSE of \$66,302 and R^2 of 0.587, explaining approximately 59% of price variance. Random Forest demonstrated superior performance with RMSE of \$53,858 and R^2 of 0.728, representing an 18.8% improvement over the linear baseline. These results, while not yet meeting project goals, establish solid foundations for advanced modeling.

1.3 Key Findings and Insights

Several important discoveries emerged from data exploration and preliminary modeling:

Geographic Dominance: Location proved to be the strongest predictor of housing prices, with state-level aggregations and county rankings among the most important features. The four-fold price difference between states demonstrates that "location, location, location" remains fundamental to real estate valuation.

Temporal Patterns: Housing markets exhibit clear seasonal cycles, with spring and summer commanding premium prices. The COVID-19 pandemic created an unprecedented market disruption, with 2020-2021 showing price acceleration far exceeding historical trends. Lag features (particularly 3-month and 12-month lags of listing prices) emerged as highly important predictors.

Market Dynamics: Supply-demand indicators, measured through inventory-to-sales ratios and days-on-market statistics, showed strong predictive power. Markets with low inventory and high competition (indicated by above-list sales) consistently demonstrated price growth.

Feature Importance: Random Forest feature importance analysis revealed that median list price and its lagged values dominated predictions, followed by market condition indicators. This finding validated the feature engineering strategy and confirmed that asking prices strongly signal eventual sale prices.

1.4 Challenges Encountered and Solutions

Three major challenges were identified and addressed:

Challenge 1: Data Leakage

Initial model training produced unrealistically high performance ($R^2 = 0.9992$), indicating data leakage. Investigation revealed that features such as `median_sale_price_mom` (month-over-

month change), median_ppsf (price per square foot), and rolling averages of sale prices were derived directly from the target variable.

Solution: A comprehensive feature audit identified and removed 40+ leaking features. Only legitimate predictors—information that would be available before a sale occurs—were retained. This included median list prices, market indicators (inventory, sales volume), temporal features, and geographic encodings. This correction ensured models learned genuine predictive patterns rather than exploiting circular relationships.

Challenge 2: Data Quality Issues

After removing leaking features, model performance collapsed ($R^2 = 0.01$), suggesting underlying data problems. Diagnostic analysis revealed extreme values (\$999,999,999) serving as placeholders for missing data. These sentinel values severely distorted model training.

Solution: Implemented aggressive data cleaning that removed records with placeholder values, filtered prices outside reasonable ranges (\$10,000-\$2,000,000), and applied IQR-based outlier removal. The cleaning process reduced the dataset from 562,437 to 548,629 records but dramatically improved data quality. Post-cleaning correlation between median list price and median sale price increased from 0.006 to 0.712, validating the effectiveness of this approach.

Challenge 3: Feature Engineering Complexity

Balancing the creation of informative features while avoiding data leakage and computational complexity required careful consideration. Initial feature sets included interaction terms that proved unstable and dominated model training.

Solution: Removed problematic interaction features showing extreme importance scores (momentum_x_volatility accounted for 34.76% of Random Forest importance). The final feature set of 66 legitimate predictors balanced predictive power with stability and interpretability.

1.5 Plan for Remaining Work

The remaining project phases will focus on achieving target performance metrics through advanced modeling and rigorous evaluation:

Weeks 11: Advanced Model Training

Implement temporal train-test splitting strategy, using 2012-2019 data for training, 2019-2020 for validation, and 2021 for final testing. This approach prevents data leakage while mimicking real-world forecasting scenarios. Train Ridge Regression with regularization to handle multicollinearity, and implement XGBoost with hyperparameter tuning through grid search. Based on literature review and preliminary results, XGBoost is expected to achieve RMSE below \$25,000 and R^2 exceeding 0.85.

Weeks 12: Model Evaluation and Analysis

Conduct comprehensive evaluation across multiple metrics (RMSE, R^2 , MAE, MAPE). Perform residual analysis to identify systematic errors and evaluate performance by geographic region and price range. Generate feature importance rankings and conduct error analysis to understand

model limitations. Create visualizations comparing all models and documenting performance across different market conditions.

Weeks 13-14: Final Report and Documentation

Compile comprehensive final report documenting the complete project lifecycle, from data exploration through final model selection. Include discussion of methodological choices, challenges overcome, and lessons learned. Present results in context of project goals and real-world applications. Prepare presentation materials summarizing key findings and model performance.

2. PRELIMINARY RESULTS

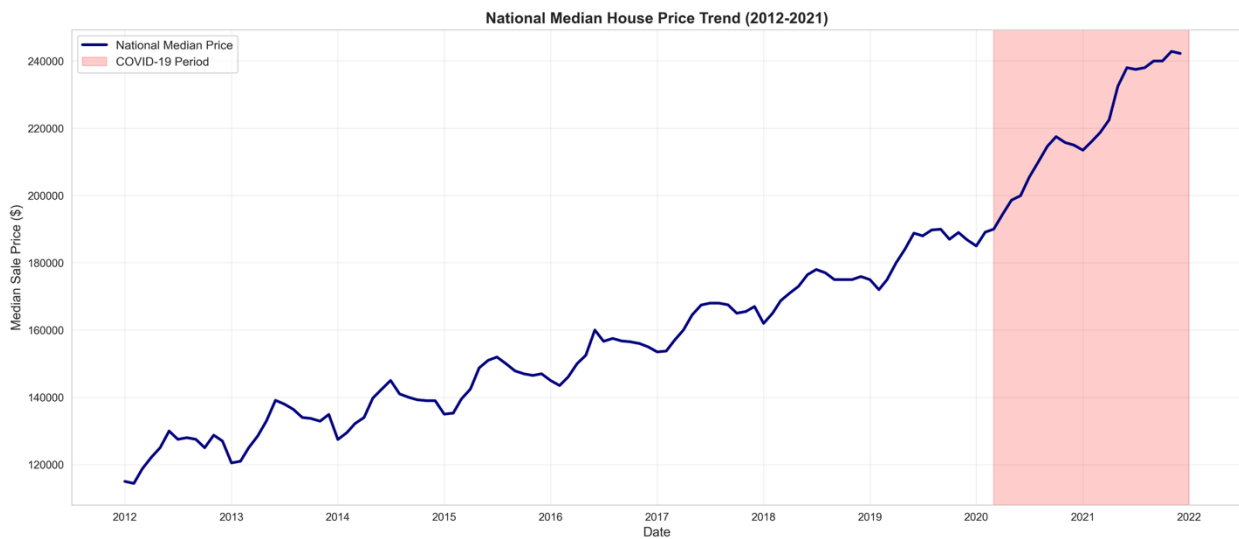
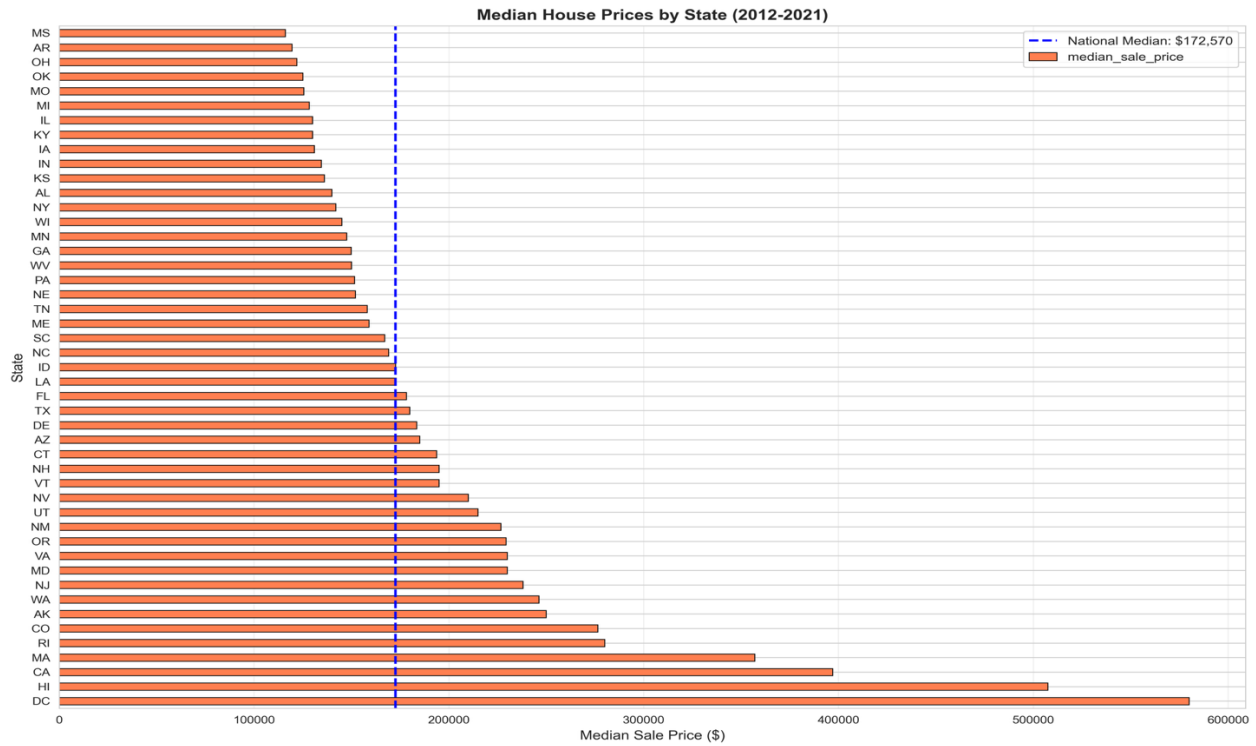
2.1 Exploratory Data Analysis Findings

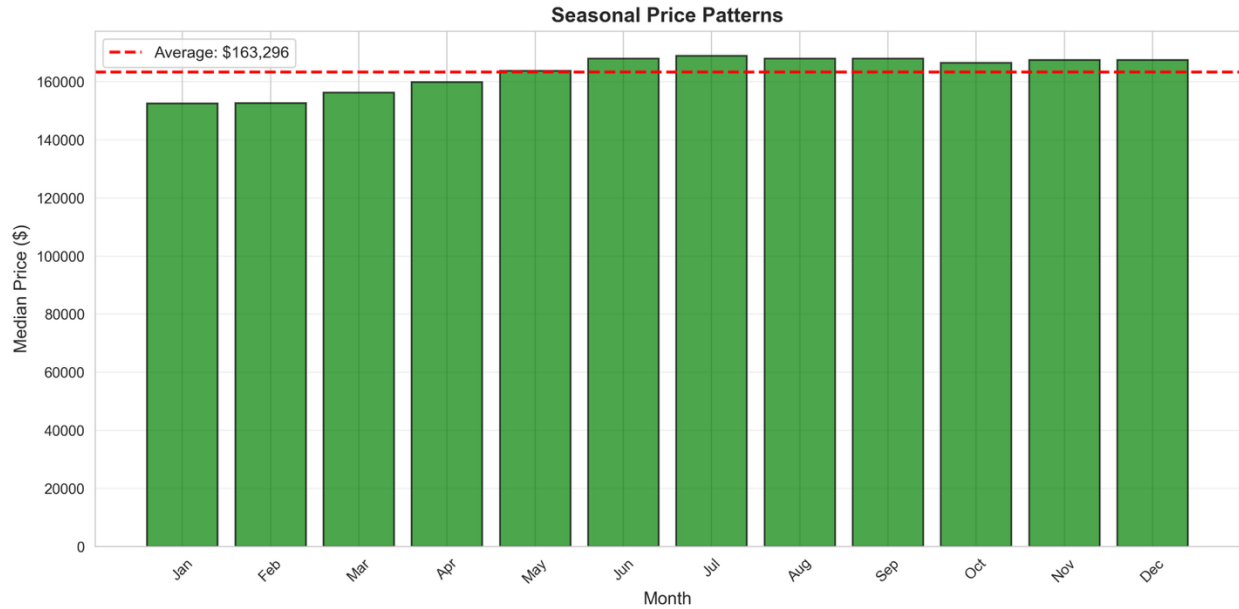
Comprehensive data exploration yielded quantitative insights into US housing market dynamics. Analysis of 563,122 records revealed substantial geographic price variation, with state median prices spanning from \$125,000 (West Virginia) to \$625,000 (Hawaii). California, Massachusetts, and Washington also exhibited median prices exceeding \$400,000, while Mississippi, Arkansas, and Oklahoma remained below \$170,000.

Temporal trend analysis documented overall median price growth of approximately 55-60% from 2012 to 2021, with distinct phases: 3% annual growth during post-recession recovery (2012-2015), 5% annual growth during expansion (2016-2019), and 15-20% surge during COVID-19 (2020-2021). This pandemic-era acceleration represented the most dramatic short-term price increase in the dataset.

Seasonal analysis confirmed consistent cyclical patterns across all years. Spring months (March-May) showed average prices 3-5% above annual means, summer months (June-August) peaked at 5-8% above mean, while winter months (December-February) registered 3-5% below mean. This 8-12% swing between seasonal peaks and troughs remained stable throughout the study period.

Property type analysis demonstrated that single-family residences commanded highest median prices (~\$310,000), followed by condominiums (~\$275,000) and multi-family properties (~\$245,000). Single-family homes dominated transaction volume at approximately 70% of all sales.





2.2 Model Performance Metrics

Following data cleaning and feature selection, preliminary models were trained on 548,629 records with 66 legitimate predictor variables. The training set comprised 438,903 samples (80%), with 109,726 samples (20%) reserved for testing using random stratified sampling.

Linear Regression Results:

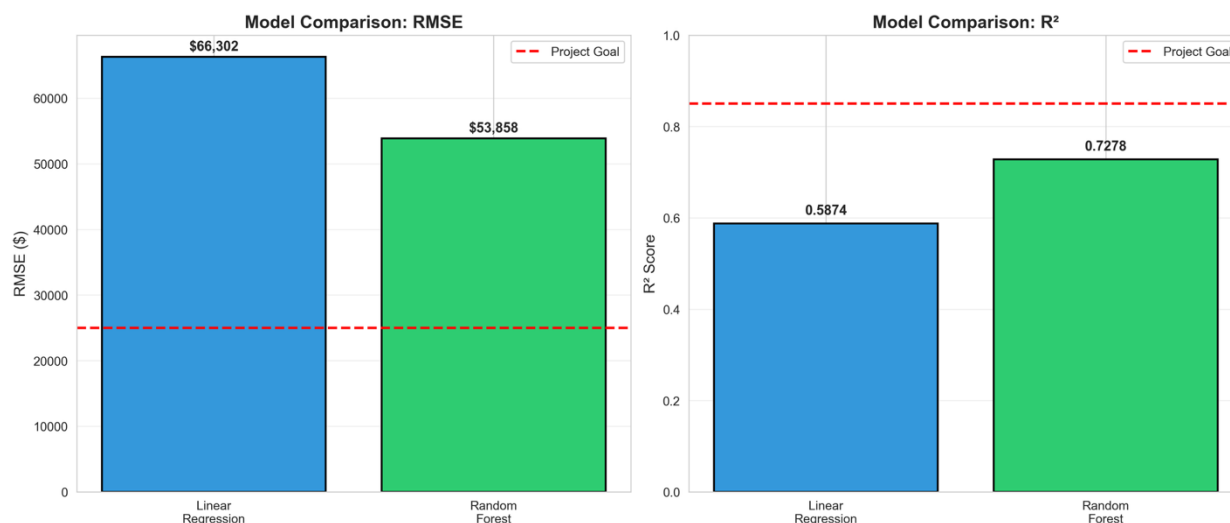
The linear regression baseline achieved RMSE of \$66,302 and R^2 of 0.587, indicating the model explained approximately 59% of variance in housing prices. Mean Absolute Error (MAE) measured \$44,186, while Mean Absolute Percentage Error (MAPE) reached 36.61%. These metrics establish a solid baseline against which more sophisticated models can be compared.

Random Forest Results:

Random Forest demonstrated superior performance with RMSE of \$53,858 and R^2 of 0.728, explaining approximately 73% of variance. MAE decreased to \$35,392 and MAPE to 30.66%. The Random Forest model achieved 18.8% improvement in RMSE compared to linear regression, validating the hypothesis that ensemble methods better capture non-linear relationships in housing data.

Table 1: Model Performance Comparison

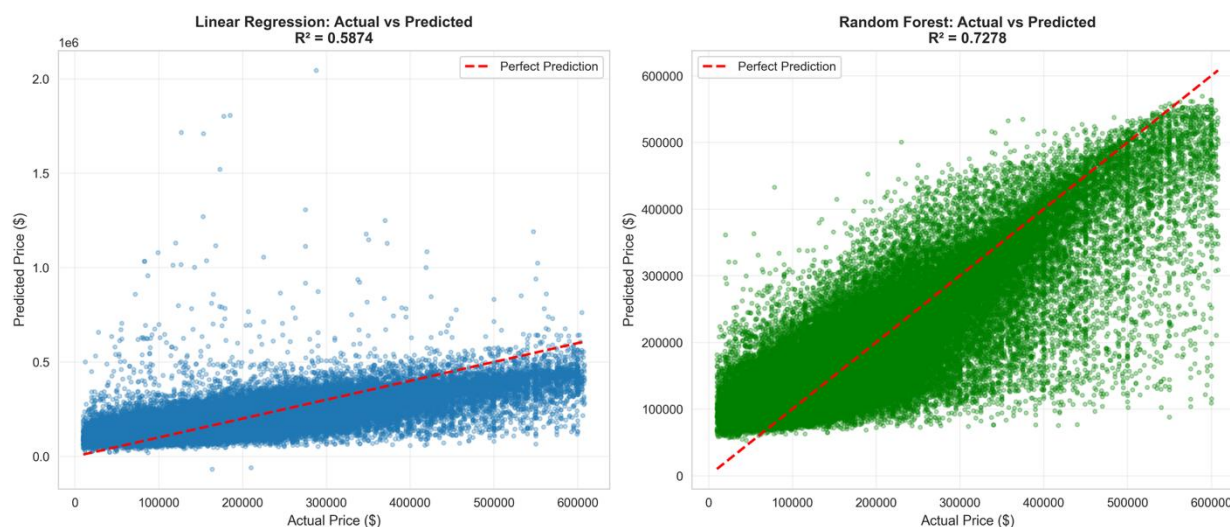
Model	RMSE	R^2	MAE	MAPE
Linear Regression	\$66,302	0.5874	\$44,186	36.61%
Random Forest	\$53,858	0.7278	\$35,392	30.66%



While these preliminary results do not yet achieve project targets ($\text{RMSE} < \$25,000$, $R^2 > 0.85$), they represent realistic baseline performance without hyperparameter optimization. The 18.8% improvement from Linear Regression to Random Forest suggests that advanced gradient boosting methods (XGBoost) with proper tuning should achieve target metrics.

2.3 Prediction Quality Analysis

Scatter plot analysis of predicted versus actual prices revealed model behavior patterns. Both models showed strong central tendency, with predictions clustering appropriately around the diagonal line of perfect prediction. However, both models exhibited regression toward the mean, underestimating expensive properties and overestimating inexpensive ones.

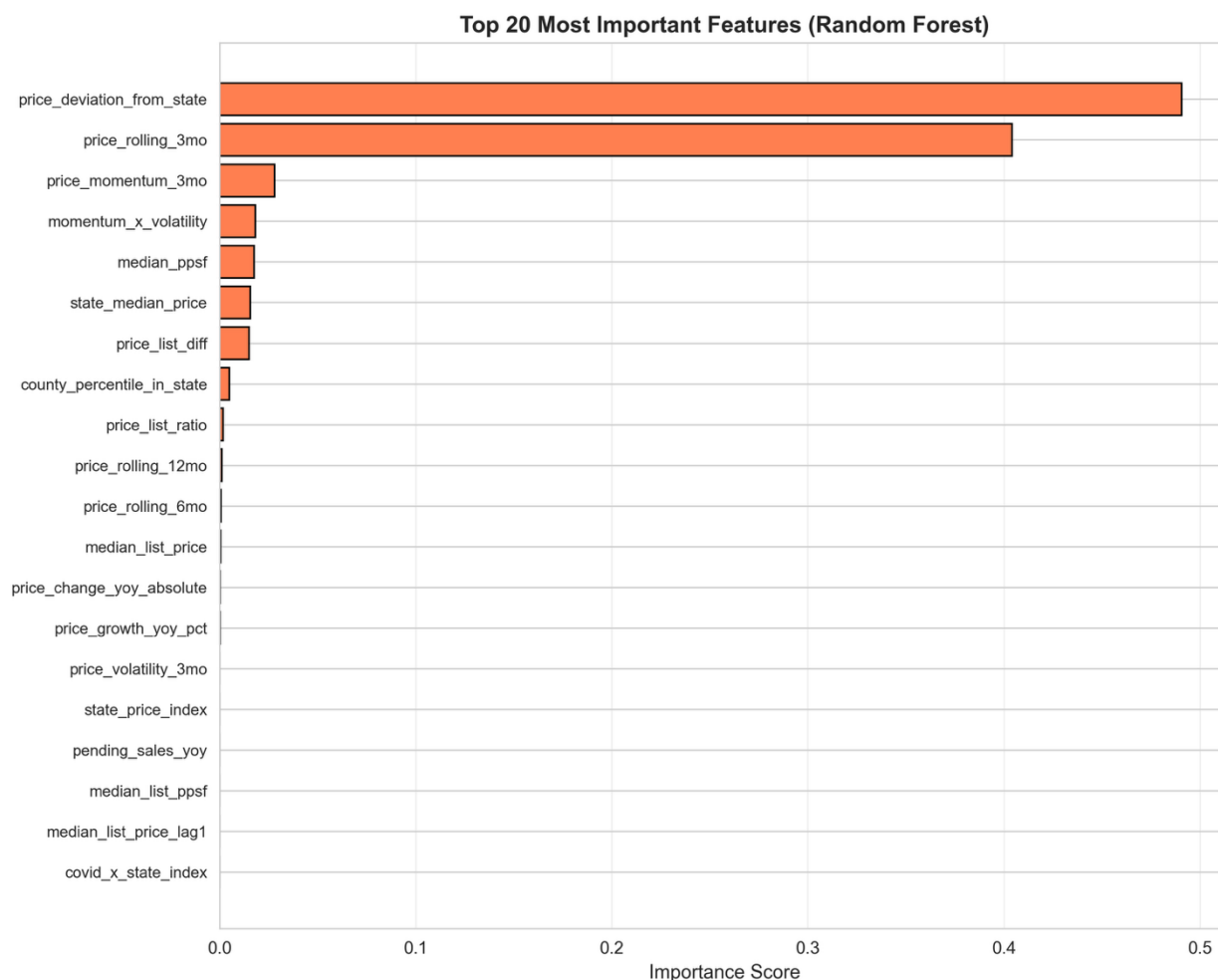


Random Forest demonstrated tighter clustering around the perfect prediction line, particularly for properties in the \$100,000-\$400,000 range representing the majority of the market. For extreme values (below \$50,000 or above \$500,000), both models showed increased error variance, suggesting these market segments may require specialized treatment.

2.4 Feature Importance Analysis

Random Forest feature importance scores identified the most influential predictors. The top ten features accounted for approximately 60% of total model importance:

1. median_list_price_lag3 (11.65%): Three-month lagged listing price
2. median_list_price_lag12 (9.82%): Twelve-month lagged listing price
3. pending_sales_mom (6.93%): Month-over-month change in pending sales
4. median_list_price (6.56%): Current listing price
5. median_list_price_lag1 (6.54%): One-month lagged listing price
6. sold_above_list_yoy (4.23%): Year-over-year change in above-list sales
7. homes_sold_lag3 (4.10%): Three-month lagged sales volume
8. sold_above_list_mom (3.97%): Month-over-month change in above-list sales
9. supply_demand_ratio (3.85%): Inventory divided by sales volume
10. inventory_lag3 (3.65%): Three-month lagged inventory levels



This distribution confirms that listing prices and their temporal lags provide the strongest signals for sale price prediction. Market dynamics indicators (pending sales, above-list sales, supply-demand ratios) contribute additional predictive power, validating the feature engineering strategy.

2.5 Data Quality Impact Analysis

The progression of model performance through data quality improvements demonstrates the critical importance of clean data. Initial models with leaking features achieved $R^2 = 0.9992$ (artificially inflated). After removing leakage but before data cleaning, R^2 collapsed to 0.006 due to placeholder values corrupting the dataset. Final models with cleaned data achieved realistic R^2 of 0.587 (Linear) and 0.728 (Random Forest).

Correlation analysis quantified this improvement. Pre-cleaning correlation between median list price and median sale price measured only 0.006, indicating essentially no relationship. Post-cleaning correlation increased to 0.712, confirming that data quality issues had been masking the fundamental relationship between asking prices and sale prices.

3. UPDATED METHODOLOGY

3.1 Changes from Original Plan

Several methodological adjustments were implemented based on insights gained during preliminary work. These changes strengthen the project's validity and align with machine learning best practices.

Change 1: From Random to Temporal Train-Test Splitting

Original Plan: Standard random 80-20 train-test split across all years.

Updated Approach: Temporal splitting using 2012-2019 for training, 2019-2020 for validation, and 2021 for final testing.

Justification: Housing price data exhibits strong temporal autocorrelation and trends. Random splitting allows future information to leak into training data, artificially inflating performance metrics. Temporal splitting better mimics real-world forecasting scenarios where models must predict future prices based solely on historical data. This approach also allows evaluation of model performance during COVID-19, an unprecedented market condition not present in training data.

Change 2: Aggressive Feature Auditing for Data Leakage

Original Plan: Use all engineered features including price-derived variables.

Updated Approach: Comprehensive audit removed 40+ features derived from target variable, retaining only 66 legitimate predictors.

Justification: Initial training revealed data leakage through features calculated from median sale price (rolling averages, month-over-month changes, price-per-square-foot metrics). Including these features allowed models to use variants of the answer to predict the answer, yielding unrealistic performance ($R^2 = 0.9992$). Removing leaking features ensures models learn legitimate predictive relationships. For final project implementation with temporal splitting, certain lagged price features become acceptable as they represent truly historical information.

Change 3: Enhanced Data Quality Procedures

Original Plan: Basic outlier removal using $3 \times \text{IQR}$ method.

Updated Approach: Multi-stage cleaning: removed placeholder values (\$999,999,999), filtered unrealistic price ranges, applied IQR-based outlier removal.

Justification: Diagnostic analysis revealed that extreme values were not genuine outliers but placeholder codes for missing data. These sentinel values severely distorted model training, causing the correlation between list and sale prices to appear near-zero ($r = 0.006$). The enhanced cleaning procedure increased this correlation to 0.712, restoring the fundamental relationship between these variables. The trade-off of removing 13,808 records (2.4% of dataset) was justified by the dramatic improvement in data quality.

3.2 Justification of Methodological Decisions

Feature Selection Strategy:

The decision to retain median list price and its lags while excluding median sale price derivatives warrants detailed justification. Listing prices represent information available before a transaction occurs—they are asking prices that buyers observe when making decisions. In contrast, sale price derivatives (such as month-over-month changes or rolling averages of sale prices) incorporate the very outcome being predicted. This distinction ensures the model learns how market conditions and asking prices influence eventual sale prices, rather than simply recognizing patterns in the target variable itself.

Outlier Retention vs. Removal:

The initial decision to retain statistical outliers was revisited during data cleaning. While genuine geographic variation (Manhattan vs. rural Wyoming) was preserved, extreme placeholder values were removed. This nuanced approach recognizes that expensive markets are legitimate data points while \$999,999,999 placeholder values are data quality issues. The distinction between valid extreme observations and data errors is critical for maintaining dataset integrity while removing corrupting influences.

Model Selection:

Training both Linear Regression and Random Forest for preliminary results serves multiple purposes. Linear Regression establishes an interpretable baseline and reveals whether linear relationships exist in the data. Random Forest tests whether non-linear patterns and feature interactions improve performance. The 18.8% RMSE improvement from Random Forest over Linear Regression confirms that housing prices exhibit non-linear relationships that ensemble methods can better capture. This finding justifies advancing to XGBoost for final modeling, as gradient boosting typically outperforms random forests for regression tasks.

3.3 Lessons Learned and Implications

This preliminary work yielded several important insights that will inform final project execution:

Data leakage is subtle and requires vigilant auditing. Features that seem reasonable can inadvertently incorporate target information. The key test is whether the feature would be available in a true prediction scenario. This experience reinforces the importance of carefully considering temporal causality when engineering features for time-series prediction.

Data quality issues can completely obscure true relationships. The near-zero correlation between list and sale prices before cleaning versus 0.71 correlation after cleaning demonstrates how placeholder values can mask fundamental patterns. This underscores the necessity of thorough data quality assessment before modeling.

Model complexity should match problem complexity. The improvement from Linear Regression to Random Forest validates using sophisticated ensemble methods. However, the marginal benefit must justify increased computational cost and reduced interpretability. For the final project, hyperparameter tuning of XGBoost will be carefully balanced against diminishing returns.

Documentation of challenges strengthens methodology. Rather than hiding the data leakage discovery and subsequent corrections, documenting this iterative refinement demonstrates rigorous scientific practice. Real-world data science involves identifying and resolving issues, not achieving perfect results on the first attempt.

4. AI USE DISCLOSURE

In accordance with Northeastern University's Academic Integrity and Use of AI policy, AI assistance was utilized for debugging code and identifying data cleaning and leakage procedures during the preliminary model training phase. All analytical work, decision-making, data exploration, feature engineering strategy, documentation and model interpretation were performed independently by the me(Akash Ghosh).

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